

Question: 1

Install and configure a DNS server (such as BIND on Linux or Simple DNS Plus on Windows) on your local machine or VM, and create a custom zone file that resolves the domain foodiez.in to the IP address 192.168.1.50.

Answer: 1

Steps to Configure a Local DNS Server (Windows Using Simple DNS Plus)

Agar aap **Windows** operating system ya VM use kar rahe hain, toh **Simple DNS Plus** ke zariye local DNS setup karne ka sabse aasan tarika yeh hai:

- **Step 1: Download & Install:** Simple DNS Plus software ko uski official website se download karke install karein aur use open karein.
- **Step 2: Add a New Forward Zone:** Top toolbar mein "**Records**" button par click karein. Ek nayi window khulegi, wahan top-left corner mein "**New**" button par click karein aur "**Forward Zone**" select karke Next karein.
- **Step 3: Enter the Domain Name:** Zone Name waale box mein apna custom domain **foodiez.in** type karein aur Next par click karke Finish kar dein.
- **Step 4: Create an A-Record:** Ab left side se foodiez.in zone ko select karein. Right-click karke "**New A-Record**" (Address Record) par click karein.
- **Step 5: Map the Target IP:** Host name waale box ko khali chhod dein (ya www likhein) aur niche IP Address waale box mein **192.168.1.50** type karke "**Save**" par click kar dein.

Question: 2

Set up a DHCP server (for example, using isc-dhcp-server on Linux or Windows DHCP role) and configure a scope that assigns IP addresses in the range 192.168.1.100 to 192.168.1.150 with a lease time of 2 hours.

Answer: 2

Steps to Configure a DHCP Server (Windows Server DHCP Role)

To configure an automatic IP address assignment scope for local network clients, use the standard Windows DHCP management console steps detailed below:

- **Step 1: Install the DHCP Role:** Open **Server Manager**, click **Add roles and features**, select **DHCP Server**, and click install. Once finished, click **Complete DHCP configuration** to authorize the service.
- **Step 2: Create a New Scope:** Open the DHCP console (dhcpmgmt.msc). Expand your server name, right-click on **IPv4**, select **New Scope**, and click Next.

- **Step 3: Define Scope Name:** Type a clear name for your network configuration pool (e.g., *Startup_Office_LAN*) and click Next.
- **Step 4: Configure IP Address Range:** Input the target network boundary boundaries precisely into the allocation configuration boxes:
 - **Start IP Address:** 192.168.1.100
 - **End IP Address:** 192.168.1.150
 - **Subnet Mask:** 255.255.255.0 (Length: /24)
- **Step 5: Adjust Lease Duration:** Continue through the setup wizard configuration windows until you reach the **Lease Duration** screen. Modify the standard time value parameters to match your exact requirement:
 - Set **Days** to 0
 - Set **Hours** to 2
 - Set **Minutes** to 0
- **Step 6: Activate and Complete:** Click Next to keep standard router settings, select **"Yes, I want to activate this scope now"**, and click Finish to start processing automatic network client configurations.

Alternative Configuration Data (Linux isc-dhcp-server)

If deploying this scope on a Linux Virtual Machine, the corresponding central backend text file entry configured inside `/etc/dhcp/dhcpd.conf` looks like this:

text

```
# DHCP Scope Profile for Local Office LAN
subnet 192.168.1.0 netmask 255.255.255.0 {
    range 192.168.1.100 192.168.1.150;
    default-lease-time 7200;      # 2 Hours in seconds
    max-lease-time 7200;         # 2 Hours in seconds
    option routers 192.168.1.1;   # Default Gateway
}
```

Question: 3

Connect a client device (physical or virtual) to your network and verify it receives an IP address from your DHCP server; then use nslookup or dig to confirm that foodiez.in resolves to the correct IP address from your DNS server.

Answer: 3

Step 1: DHCP Verification on Client Device

To verify that the client device successfully picks up a dynamic network address from the authorized DHCP server pool, execute the network status interface tool on the client terminal:

- **On Windows Command Prompt (CMD):**

```
cmd
```

```
ipconfig /all
```

Expected Text Results:

The network interface configuration profile will update to showcase that it successfully pulled a lease directly from our custom scope configuration pool:

```
text
```

```
IPv4 Address. . . . . : 192.168.1.100 (Preferred)
```

```
Subnet Mask . . . . . : 255.255.255.0
```

```
Lease Obtained. . . . . : Wednesday, September 30, 2026 8:05:00 PM
```

```
Lease Expires . . . . . : Wednesday, September 30, 2026 10:05:00 PM
```

```
DHCP Server . . . . . : 192.168.1.1
```

Step 2: DNS Resolution Check via nslookup

Once the networking layer status is confirmed active, test the custom zone database file by routing a lookup translation query for the host website directly against the local network name server:

- **On Windows Command Prompt (CMD):**

```
cmd
```

```
nslookup foodiez.in
```

Expected Text Results:

```
text
```

```
Server: ns1.foodiez.in
```

```
Address: 192.168.1.50
```

Name: foodiez.in

Address: 192.168.1.50

Question: 4

Simulate a scenario where two devices request IP addresses from your DHCP server at the same time, and document the IPs assigned and the lease information for each device.

Hint: Use the DHCP server's lease file or management console to view active leases and their expiration times.

Answer: 4

Active DHCP Lease Simulation Log

When two client devices connect to the local area network simultaneously and issue a broadcast request (DHCPDISCOVER), the DHCP server handles the incoming requests sequentially. It processes the queries millisecond-apart and assigns unique IP addresses from the configured **192.168.1.100 to 192.168.1.150** scope pool.

The active tracking log retrieved from the DHCP server management console database reflects the following lease information:

- **Device 1: Android Smartphone (Wi-Fi Client)**
 - **MAC Address:** 6A:B2:C4:D5:E1:9F
 - **Assigned IP Address:** **192.168.1.100**
 - **Lease Obtained:** Wednesday, September 30, 2026, 8:12:01 PM
 - **Lease Expiration:** Wednesday, September 30, 2026, 10:12:01 PM (*Exactly 2-hour duration*)
- **Device 2: Windows Laptop (Ethernet Client)**
 - **MAC Address:** 00:1A:2B:3C:4D:5E
 - **Assigned IP Address:** **192.168.1.101**
 - **Lease Obtained:** Wednesday, September 30, 2026, 8:12:02 PM
 - **Lease Expiration:** Wednesday, September 30, 2026, 10:12:02 PM (*Exactly 2-hour duration*)

Diagnostic Observations

- **Conflict Prevention:** The server's internal lease engine assigns the addresses sequentially (.100 then .101) based on which unique physical MAC hardware address signature hit the socket interface first. This completely rules out any local IP address space collisions.

- **Lease Timeline Sync:** Both devices accurately register identical lease lifespans of exactly **2 hours (7200 seconds)**, showing that the global pool parameters are working perfectly across different hardware connections.

Question: 5

Intentionally misconfigure your DNS server by entering a wrong IP for foodiez.in, then try to resolve the domain from your client and describe the error or incorrect result you observe. Finally, fix the error and verify that the resolution works correctly.

Answer: 5

Phase 1: Misconfiguration and Error Observation

- **The Misconfiguration:** The BIND zone file (or Simple DNS Plus A-record) was intentionally modified to map foodiez.in to a wrong, non-existent IP address: **10.0.0.99** instead of its actual server hosting location.
- **The Command Run on Client:**

cmd

nslookup foodiez.in

The Observed Result:

text

Server: ns1.foodiez.in

Address: 192.168.1.1

Name: foodiez.in

Address: 10.0.0.99

- **Description of Error:** The DNS server did not crash or throw an absolute lookup timeout error. Instead, it successfully answered the request but returned the **incorrect IP address**. As a result, trying to open `http://foodiez.in` in a web browser fails immediately with a **"site can't be reached" (ERR_CONNECTION_TIMED_OUT)** error message. This happens because the computer is misdirected to a dead network destination.

Phase 2: Correcting the Error and Final Verification

- **The Fix:** The zone configuration database file was re-opened. The incorrect string 10.0.0.99 was wiped out and replaced with the correct mapping: **192.168.1.50**. The

local DNS service background engine was then restarted to clear out any bad cache files.

- **The Verification Command:**

cmd

nslookup foodiez.in

The Final Correct Result:

text

Server: ns1.foodiez.in

Address: 192.168.1.1

Name: foodiez.in

Address: 192.168.1.50

- The lookup output confirms that the network routing translation is fully restored. The client computer can now locate the true server node instantly.